

① Three ways to define relation:

• informal: table of values

← attribute type or column				
	← tuple or row			
		← cell or value		

• bottom-up: set of tuples

- values in each tuple come from the domains of the columns
i.e. attribute types

• top-down: subset of Cartesian product of column domains

② Definition of relation schema (not in textbook):

• Collection of attribute types with some constraints on the permitted values of the tuples

- the most common constraint is that a certain column must contain unique values e.g. Student ID number
- or sometimes a collection of column values must be unique e.g. course num AND section number

③ Definition of superkey:

any collection of columns whose values are guaranteed to be unique

from here on,
A, B, C, ... are columns
(i.e. attribute types)

- some of the values could be repeated, but every pair of tuples differs in at least one of these columns e.g.
- if A is a superkey, so is AB
- if AB is a superkey, so is ABC

Course num C	Section num S	Instructor I
378	1	JM
378	2	JM
314	1	JM

CS is a superkey
CSI is a superkey
C is not a superkey
S is not a superkey

④ Definition of key:

- a minimal superkey
 - if you delete any column from the key, it's no longer a superkey
 - if A is a superkey, then it's a key.
 - if AB is a superkey, and A is not a key, and B is not a key, then AB is a key

e.g. CS above is a key
CSI is a superkey but not a key.

⑤ Definition of candidate key:

- same thing as a key
- just emphasizes that there can be more than one

⑥ Definition of primary key:

- the database designer chooses one key to be the primary key (PK) of a relation
- the PK could be a single column or multiple columns.

⑦ Definition of alternate key: any key that isn't the PK.

⑧ Definition of foreign key:

We have 2 relations, R and S.

A is a col in R

B is a col in S. B is the primary key in S.

A is a foreign key (FK) referring to B in S
if every value in A must appear in B.

• A and B could be collections of columns.

• the FK links every row of R to some row of S

e.g. R (courses)

num	instr-id
378	20
314	41
132	15

↑ FK referring to id in S

id	initials
15	FS
20	LF
41	MS
37	GB
20	JM

phone	office

↑ PK

S (instructors)

id	initials	phone	office	etc...
15	FS			
20	LF			
41	MS			
37	GB			
20	JM			

↑ PK

⑨ Relation notation (this term, "relation notation", is not used in the textbook but they use the notation - see e.g. § 61.5)

• We describe database relations using relation notation in the format

RelationName (ColName1, ColName2, ...)

• The PK is underlined e.g. Instructor (id, initials, phone, office)

• FKs are in italics e.g. Course (num, *instr-id*)
with separate explanation of what the FK refers to
FK instr-id refers to id in Instructor.